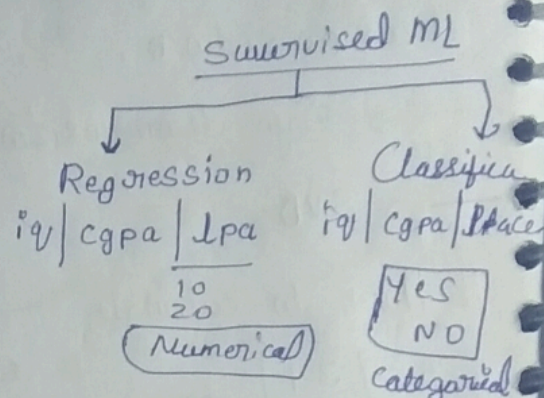


Model Evaluation & Selection

Why ROC Curve:-

[Threshold value] $\rightarrow$ classification

iq	cgpa	placed	} 1000 data	
-	-	-		
			training	test
			500	200
iq	cgpa	placed	Prediction	Pred-Prob
7	70	1	0	0.45 ✓
8	80	0	0	0.39 ✗
9	90	1	1	0.61 ✓



Considering:-

threshold ≥ 0.5

what is threshold value:-

A Threshold value is a cut-off value used to convert probability into a class.

$\rightarrow$ Changing threshold values changes model predictions, Even if the model is same.

why threshold creates a problem:-

High threshold

- $\rightarrow$ model is very strict (Ex: 0.9)
- $\rightarrow$ Predicts positive only when very sure.
- $\rightarrow$ Few positives detected
- $\rightarrow$ Few False alarms

Low threshold

- $\rightarrow$ model is very lenient (0.35)
- $\rightarrow$ Predicts positive for many cases
- $\rightarrow$ many positives detected
- $\rightarrow$ many false alarms.

what ROC wants to study:-

- $\rightarrow$ ROC does not want :- which threshold is best?
- $\rightarrow$ ROC wants :- "How does the model behavior change as the threshold moves."

- So ROC tries $\Rightarrow$
- Tries all possible thresholds
 - observes how error change
 - plots the behavior

Ex: # accuracy fails, # diseases

→ 1,000 people,

→ 990 - healthy

→ 10 - diseased

⇒ model predicts: - Everyone as "healthy"

$$\text{accuracy} \Rightarrow \frac{990}{1000} = 99\%$$

looks great but model detects zero patients

→ So accuracy ignores type of mistakes

→ Doesn't care which class is important (Consider a person having cancer, but predicted as healthy)

Why ROC: - ROC curve needed because it evaluates a classifier's ability to distinguish between classes across all possible thresholds, especially when accuracy is misleading.

ROC [Receiver Operating Characteristic]: -

ROC is a curve (graph) that shows how well a binary classification model can distinguish two classes as we change the decision threshold.

Confusion matrix:

	Predicted +ve (1)	Predicted -ve (0)
Actual +ve (1)	✓ True Positive (TP)	False Negative (FN)
Actual -ve (0)	False Positive (FP)	True Negative (TN) ✓

True Positive Rate (TPR): - Benefit

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \uparrow \text{maximize, } \text{TPR} = 1, \text{ when, FN} = 0$$

$\text{TPR} = 0, \text{ when TP} = 0$

TPR tells us: - out of all the actual positive cases, how many did the model correctly identify? +ve

→ How good is the model at catching positives?

Ex:- # accuracy fails, # diseases → 10 - diseased
 → 1,000 people, → 990 - healthy
 ⇒ model predicts:- Everyone as "healthy"
 { Looks great but model detects zero patients

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True Positive Rate (TPR):- Benefit

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \uparrow \text{maximize}$$

TPR = 1, when, FN = 0
 TPR = 0, when TP = 0

TPR tells us:- out of all the actual positive cases, how many did the model correctly identify? +ve

→ How good is the model at catching positives?

Ex:-

- 100 people tested
- 20 people actually have the disease

model prediction:-

- * correctly identified 15 diseased people

True $\Rightarrow$ 1 (disease)

False $\Rightarrow$ 0

$$(TP = 15)$$

* missed 5 diseased people $\Rightarrow$ FN (5)

$$(recall) \underline{TPR} \Rightarrow \frac{TP}{TP+FN} \Rightarrow \frac{15}{15+5} = \frac{15}{20} = 0.75$$

$\rightarrow$ The model detects 75% of the people who actually have the disease.

False Positive Rate (FPR): $\rightarrow$ Cost

FPR tells us:- out of all the actual negative cases, how many did the model wrongly classify as positive.

$$\boxed{FPR = \frac{FP}{FP+TN}} \downarrow \underline{\text{best}} \Rightarrow FPR = \frac{10}{10+70} = \frac{10}{80} = 0.125$$

Ex:- 100 people tested

- 80 are healthy (Negative)

model predicts:-

- 10 healthy people wrongly predicted as disease $\rightarrow FP = 10$
- 70 healthy people correctly predicted as healthy $\rightarrow TN = 70$

$\rightarrow$ Out of all healthy people, model falsely alarms 12.5%.

Ex:- cost:- Churn rate prediction $\Rightarrow$

who were not leaving platform (TN)

out of 100 people $\rightarrow$ (90)

$\rightarrow$ 10 people will leave (FP)

for those who are leaving, giving discount of 2000 $(100 \times 10) \Rightarrow 2000$

$$FPR = 1, \text{ when } TN = 0$$

(loss of 1000) $\Rightarrow$ because model predicted 10 people are leaving. It is False

$$\boxed{FPR = 0, \text{ when } FP = 0} \text{ best case}$$

best case $\Rightarrow$

	1	0
1	TP	FN
0	FP	TN

100	0
0	100

when misclassification is zero, that is the best case

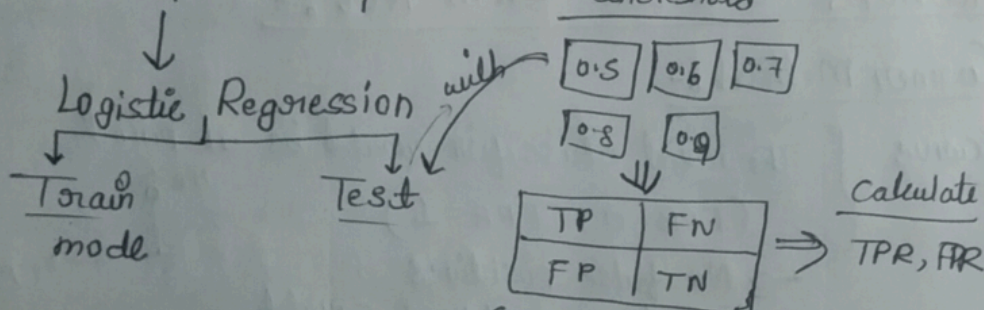
ROC Curve:-

ROC curve is a graph that shows how a classification model's True Positive Rate (TPR) and False Positive Rate (FPR) change when we move from the decision threshold from high to low.

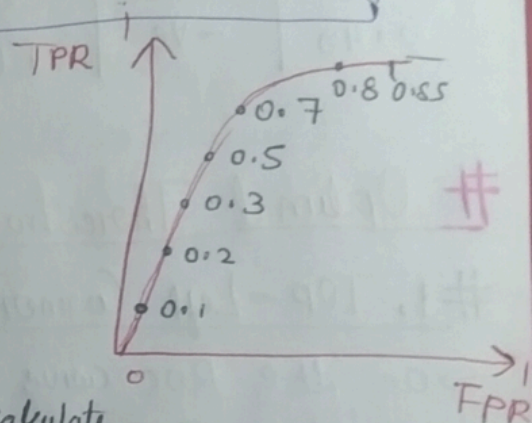
Why ROC needed:-

- model gives probabilities, not Yes/No
- Threshold decides the final class
- Changing threshold changes.
 - TPR (how many positives we catch)
 - FPR (how many false alarms we make)
- Accuracy depends on 1 threshold - misleading
- ROC shows the model behavior at all thresholds

Cgpa | iq | Placement
= | - | Yes
No



(Confusion matrix for each threshold)



i) Low Threshold (lenient)

• Predicts Positives very Easy

TP ↑	FN ↓
FP ↑	TN ↓

$$TPR = \frac{TP}{TP+FN} \Rightarrow \uparrow$$

$$FPR = \frac{FP}{FP+TN} \Rightarrow \uparrow$$



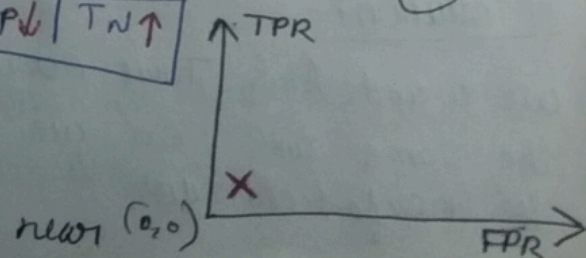
ii) High Threshold

• Predicts positive only when very confident.

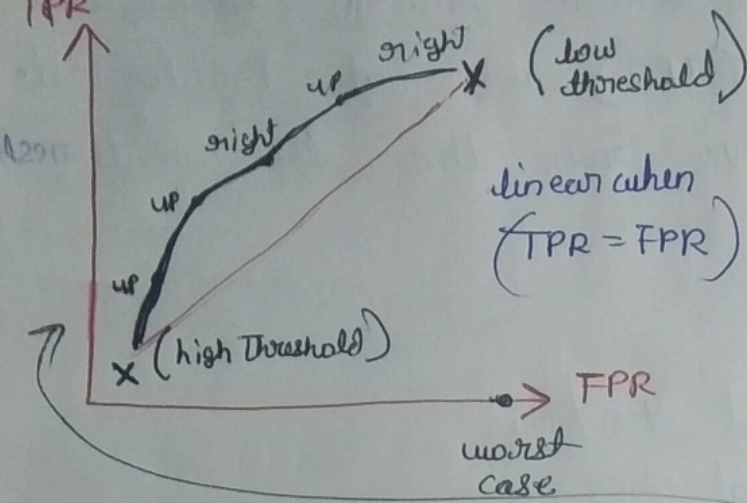
TP ↓	FN ↑
FP ↓	TN ↑

$$TPR = \frac{TP}{TP+FN} \Rightarrow \downarrow$$

$$FPR = \frac{FP}{FP+TN} \Rightarrow \downarrow$$



TPR is not proportional to FPR



Threshold = 0.8

Ex:-

	TP ↑	FP ↑	FN ↓
actual ⇒ Spam	not Spam	Spam	Spam
Predict ⇒ Spam	Spam	not Spam	not Spam

Ex:-

Probab	Actual	ROC movement	Pattern
0.95	+ve	TPR ↑	up
0.90	+ve	TPR ↑	up
0.85	-ve	FPR ↑	Right
0.60	+ve	TPR ↑	up
0.40	-ve	FPR ↑	Right

• ROC is curved because changing threshold includes samples one by one, and depending on whether next sample is +ve / -ve, only TPR / FPR changes not both

Optimal Threshold Selection from ROC :-

#1. Top-Left Corner Method :-

→ On the ROC curve

- X-axis = FPR
- Y-axis = TPR

The ideal classifier would be at point (FPR=0, TPR=1)

- No false positives
- all true positives caught

→ since no real model is perfect, we choose threshold closest to this ideal point

$$\text{distance} = \sqrt{(1 - \text{TPR})^2 + (\text{FPR})^2}$$

select threshold with maximum distance

Intuition :-

We want high True positives and low false positives at the same time, so we choose the threshold that closest to the perfect classifier

* # 2. Youden's J Statistic :-

why this method :- Instead of geometry (distance) and

- Directly measures separation power
- Combines sensitivity (TPR) & Specificity (1-FPR)

$$J = TPR - FPR$$

how it works :-

- Calculate , TPR, FPR, J
 - Choose the threshold where J is maximum
- This threshold gives : High TPR (catch positives)
Low FPR (avoid false alarms)

Intuition:-

" Youden's J finds the threshold where the gap between true positives and false positives is maximum.

- we gain most correct detections, while minimizing wrong detections.

when to use :-

- Medical diagnosis, Academic Research
- when both FP and FN matter equally

#3. Business - Defined Constraints

why this method :- Roc curve is model-centric, not business centric

in real life :-

- Errors have diff costs
- Some mistakes are unacceptable

So business rules comes before metrics,

How it works :-

→ Business define constraints,

- $fpr \leq 5\%$
- Recall $\geq 90\%$
- false positive cost more than false negative.

→ From ROC:- filter thresholds that satisfy the constraints.

→ Among remaining threshold:- Choose the one with best performance (usually max TPR or max Precision)

Ex:- Real world Explanation (FP) -

i) Fraud detection:- Blocking genuine users is costly, so we cap FPR at 5%, among those thresholds, we select the one with maximum TPR, why not FPR = 0, causes strict model
TPR ↑, FPR ↓

ii) Medical Diagnosis:- Cancer-Screening (misses many frauds)

→ If model says No cancer, but patient actually has cancer (FN). Then Patient does not get treatment

- disease progresses
- Can become life-threatening

∴ So FN are very costly.

Lower Threshold $\Rightarrow$ model says "Yes" disease more easily

This cause $\Rightarrow$ more TP (good)
more false positive (Side effect)

What is AUC-ROC:-

→ ROC $\Rightarrow$ curve of TPR vs FPR at diff thresholds

→ AUC $\Rightarrow$ area under the ^{ROC} curve

→ It gives one number to summarize the model performance

AUC tells how well the model separates positives and negative classes.

What does AUC actually mean:-

Probability that the model gives high score to a positive sample than a negative sample.

- Ex:-
- Pick one sick patient
 - Pick one healthy patient
 - If model scores high for a sick person — correct

AUC	meaning
1.0	Perfect model
0.8	good separation
0.5	Random guessing
<0.5	model is in wrong direction

why AUC instead of accuracy

- accuracy depends on threshold
- accuracy fails on imbalanced data
- AUC is threshold-independent
- AUC measures ranking quality, not final labels

when AUC is useful

- Comparing models
- when threshold is not fixed
- Binary classification
- Early model selection

when not to trust AUC alone

- Highly imbalanced data
- Business cares about specific threshold.
- Precision is more important

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